

生态研究前沿速览_2026-05-02

生态研究前沿速览

2026-05-02

共收录 40 篇最新且重要研究

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生态学核心

1. Flexible photosynthesis at the end Permian

(二叠纪末期植物光合作用策略的灵活性)

Nature Ecology & Evolution | 2026-04-20 | 【Recent】

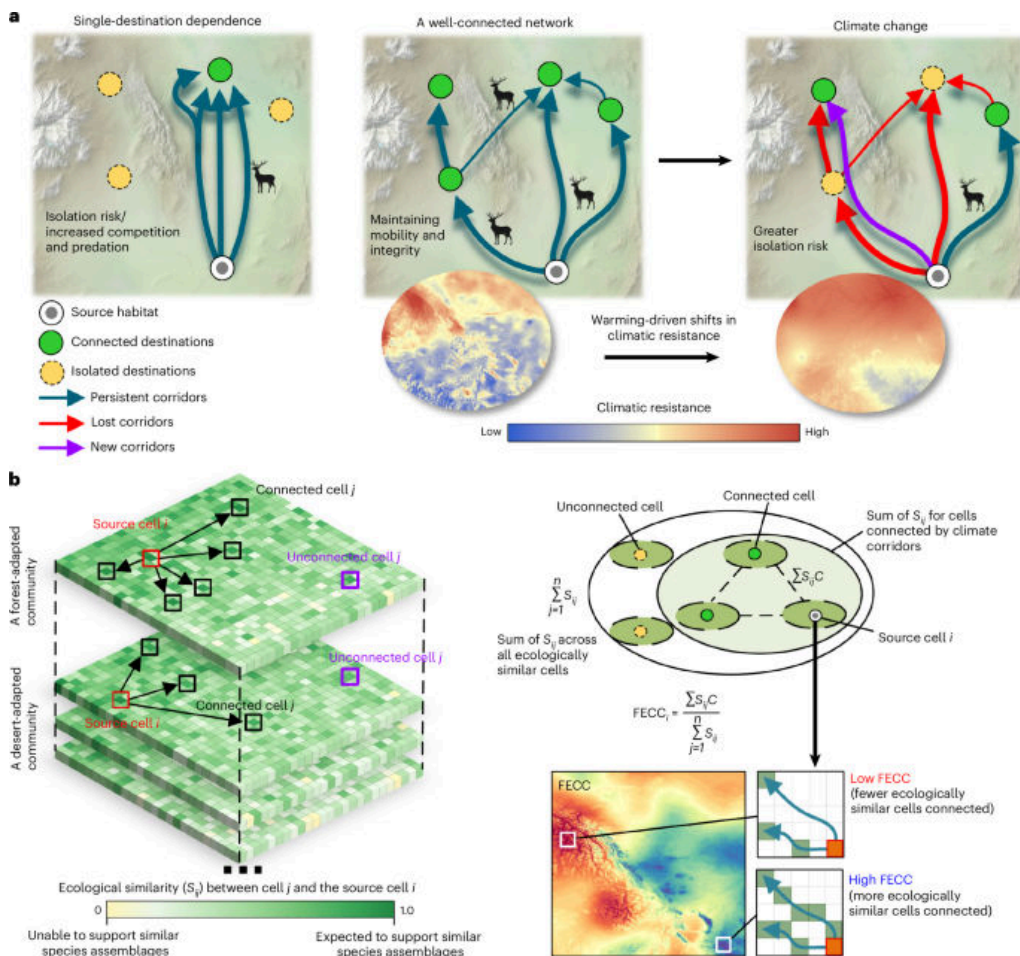


图: Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03052-y The few plants that thrived after the Earth's greatest mass extinction used a specialized physiology, known as crassulacean acid metabolism photosynthesis, that is adapted to extremely dry environments.

中文摘要

研究围绕“二叠纪末期植物光合作用策略的灵活性”展开。英文摘要要点为: The few plants that thrived after the Earth's greatest mass extinction used a specialized physiology, known as crassulacean acid metabolism photosynthesis, that is adapted to extremely dry environments. 该条目发表于 Nature Ecology & Evolution, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

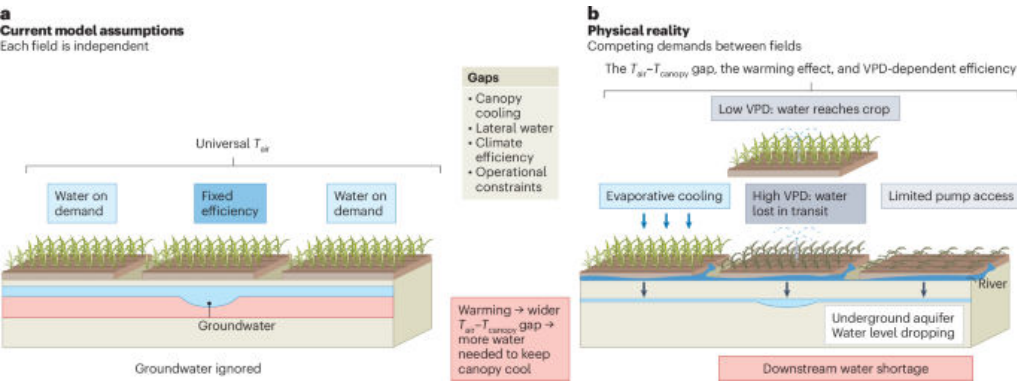
该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03052-y>

生态学核心

2. CAM photosynthesis may have conferred an advantage during the Permian–Triassic mass extinction event (CAM 光合作用可能帮助植物熬过二叠纪—三叠纪大灭绝)

Nature Ecology & Evolution | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03026-0 Morphological and stable isotope analysis of Early Triassic lycoplyte leaves suggest they were similar to extant Isoetales and thus may have made use of crassulacean acid metabolism photosynthesis—a trait that may have been advantageous during the Permian–Triassic mass extinction.

中文摘要

研究通过早三叠世石松类叶片的形态和稳定同位素分析发现，这些植物可能类似现代水韭目植物，并采用景天酸代谢（CAM）光合作用。这种更适应极端干旱与环境胁迫的生理策略，可能解释了部分植物在二叠纪—三叠纪大灭绝后仍能存活并扩张。

深度解读

这项研究把古植物学、形态证据和稳定同位素线索拼到了一起，给出了一个很有说服力的解释：并不是所有植物都被动承受大灭绝，某些类群可能靠更灵活的光合策略挺了过去。CAM 通常与干旱、高温和水分利用效率提升相关，因此它在极端气候背景下的优势非常合理。更重要的是，这篇文章提醒我们，大灭绝后的生态重建不只是‘谁死了谁活着’，还涉及生理机制层面的筛选。对理解植物如何应对未来极端气候，它也提供了深时尺度上的启示。

DOI: <https://doi.org/10.1038/s41559-026-03026-0>

3. Dynamics of island mass effect –Part 2: Phytoplankton physiological responses (岛屿质量效应动态之二：浮游植物的生理响应)

Biogeosciences | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Dynamics of island mass effect – Part 2: Phytoplankton physiological responses Guillaume Bourdin, Lee Karp-Boss, Fabien Lombard, Gabriel Gorsky, and Emmanuel Boss Biogeosciences, 23, 2687 – 2728, <https://doi.org/10.5194/bg-23-2687-2026>, 2026 Island mass effect (IME) refers to elevated chlorophyll a concentrations around islands, often extending hundreds of kilometers into oligotrophic waters. Using satellite-derived nutrient and iron stress indices and in situ bio-optical data from the Tara Pacific expedition, we show that IMEs are associated with recurring iron and macronutrient enrichments, and changes in phytoplankton community composition relative to the regional background ocean.

中文摘要

研究围绕“岛屿质量效应动态之二：浮游植物的生理响应”展开。英文摘要要点为：Dynamics of island mass effect –Part 2: Phytoplankton physiological responses Guillaume Bourdin, Lee Karp-Boss, Fabien Lombard, Gabriel Gorsky, and Emmanuel Boss Biogeosciences, 23, 2687–2728, <https://doi.org/10.5194/bg-23-2687-2026>, 2026 Island mass effect (IME) refers to elevated chlorophyll a concentrations around islands, often extending hundreds of kilometers into oligotrophic waters. Using satellite-derived nutrient and iron stress indices and in situ bio-optical data from the Tara Pacific expedition, we show that IMEs are associated with recurring iron and macronutrient enrichments, and changes in phytoplankton community composition relative to the regional background ocean. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2687-2026>

综合顶刊

4. Historical legacies shape continental variation in contemporary mammal food webs

(历史遗产塑造当代哺乳动物食物网的大陆差异)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Significance Trophic interactions underpin ecosystem structure and function, yet how past species losses, evolutionary histories, and environmental variability shape macroecological patterns of food webs remains poorly understood. We present continental-

中文摘要

研究指出，营养相互作用支撑生态系统结构与功能，但过去物种丧失、演化历史和环境变异如何共同塑造大陆尺度食物网格局仍缺乏清晰认识。作者从大陆尺度比较当代哺乳动物食物网，强调历史遗产会影响今天不同区域食物网的组成、连接方式和功能结构。

深度解读

这篇论文的重要性在于把历史灭绝和当代食物网结构联系起来，避免把当前生态网络视为静态或完全由现时环境决定。大型哺乳动物和顶级消费者的历史丧失会改变营养级结构，并影响生态系统功能恢复的上限。对保护规划而言，恢复食物网需要考虑区域历史基线，而不仅是现存物种清单。该研究也为再野化和大型动物恢复提供了更宏观的比较框架。

DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2519938123>

环境与气候

5. Turning the tide of climate solutions to save tidal flats

(扭转气候解决方案方向以拯救潮滩)

Nature Sustainability | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 20 April 2026; doi:10.1038/s41893-026-01802-1 Tidal flats are vital carbon sinks and biodiversity hotspots, yet they face increasing threats from misapplied climate solutions, including mangrove afforestation and renewable energy expansion. Protecting tidal flats from well-intentioned but misapplied solutions is crucial for aligning global climate and biodiversity goals.

中文摘要

研究围绕“扭转气候解决方案方向以拯救潮滩”展开。英文摘要要点为：Tidal flats are vital carbon sinks and biodiversity hotspots, yet they face increasing threats from misapplied climate solutions, including mangrove afforestation and renewable energy expansion. Protecting tidal flats from well-intentioned but misapplied solutions is crucial for aligning global climate and biodiversity goals. 该条目发表于 Nature Sustainability，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41893-026-01802-1>

综合顶刊

6. Leveraging remote sensing and crowd-sourced biodiversity data for enhanced plant functional trait mapping (融合遥感与众包生物多样性数据可提升全球植物功能性状制图)

Nature Communications | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Nature Communications, Published online: 21 April 2026; doi:10.1038/s41467-026-72111-6 Plant functional traits govern ecosystem processes but lack spatially detailed global representations. Here, the authors combine remote sensing and crowd-sourced biodiversity data to map leaf trait distributions and their higher-order moments worldwide.

中文摘要

研究结合遥感观测与众包生物多样性数据，绘制了全球尺度叶片功能性状及其高阶统计矩分布图。该方法显著提升了植物功能性状的空间表达能力，为理解生态系统功能和全球变化响应提供了更精细的数据基础。

深度解读

植物功能性状一直是连接物种组成与生态系统过程的关键桥梁，但过去最大的问题是空间分辨率不够。本文把遥感和大规模众包生物多样性数据结合起来，相当于给全球植物性状地图补上了‘细节层’。这不仅有助于提高碳循环、植被动态和生态系统服务模型的真实性和准确性，也说明公民科学数据在全球生态监测中的价值越来越高。对未来生态预测来说，‘有性状但没地图’这个老短板正在被补上。

DOI: <https://doi.org/10.1038/s41467-026-72111-6>

土壤与生物地球化学

7. Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters

(利用光学与生态代理指标量化贫营养水体浮游植物碳)

Biogeosciences | 2026-04-20 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters David Antoine, Chandanlal Parida, and Camille Grimaldi Biogeosciences, 23, 2641 – 2660, <https://doi.org/10.5194/bg-23-2641-2026>, 2026 A dataset of phytoplankton cell counts, pigments, particulate organic carbon and optical properties enables comparison of three methods to estimate phytoplankton carbon (C_{phyto}) in oligotrophic waters, where uncertainties in phytoplankton productivity are still large. Two methods based on chlorophyll concentration and particulate backscattering, are scalable to global scale while cell counts reduce bias from non-algal material. This comparison clarifies uncertainties in optical C_{phyto} estimates.

中文摘要

研究围绕“利用光学与生态代理指标量化贫营养水体浮游植物碳”展开。英文摘要要点为：Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters David Antoine, Chandanlal Parida, and Camille Grimaldi Biogeosciences, 23, 2641 – 2660, <https://doi.org/10.5194/bg-23-2641-2026>, 2026 A dataset of phytoplankton cell counts, pigments, particulate organic carbon and optical properties enables comparison of three methods to estimate phytoplankton carbon (C_{phyto}) in oligotrophic waters, where uncertainties in phytoplankton productivity are still large. Two methods based on chlorophyll concentration and particulate backscattering, are scalable to global scale while cell counts reduce bias from non-algal material. This comparison clarifies uncertainties in optical C_{phyto} estimates. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

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DOI: <https://doi.org/10.5194/bg-23-2641-2026>

土壤与生物地球化学

8. Soil moisture-induced changes in land carbon sink projections in CMIP6

(土壤水分过程会显著改变 CMIP6 对陆地碳汇的未来预测)

Biogeosciences | 2026-04-21 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Soil moisture-induced changes in land carbon sink projections in CMIP6 Lea Maria Gabele, Petra Sieber, Laibao Liu, and Sonia I. Seneviratne Biogeosciences, 23, 2729–2746, <https://doi.org/10.5194/bg-23-2729-2026>, 2026 We investigate how soil moisture influences the future land carbon sink using dedicated experiments from the latest generation of Earth system models. Our results show that soil moisture strongly reduces land carbon uptake, mainly through soil moisture–atmosphere coupling. However, models disagree on the magnitude of SM impacts. The study emphasises that improving the representation of plant water stress in models is essential to reduce uncertainties in carbon cycle projections.

中文摘要

研究利用新一代地球系统模型专门试验，评估土壤水分如何影响未来陆地碳汇。结果显示，土壤水分主要通过土壤—大气耦合显著抑制陆地碳吸收，但不同模型对影响强度差异很大，说明植物水分胁迫表征仍是碳循环预测中的关键不确定源。

深度解读

陆地碳汇预测常常把注意力放在二氧化碳浓度和温度上，但这篇文章提醒我们，土壤水分可能是更硬的限制条件。它不仅影响植物生长，还会通过陆气耦合放大或重塑整个碳吸收过程。更麻烦的是，不同模型在这一点上分歧很大，说明目前的未来碳汇判断还带着明显的结构性不确定性。对气候目标评估而言，这意味着我们不能对陆地持续吸碳过于乐观。

DOI: <https://doi.org/10.5194/bg-23-2729-2026>

土壤与生物地球化学

9. Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment (技术札记：海洋沉积物中海藻酸盐提取与比色分析方法)

Biogeosciences | 2026-04-21 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment Sota Nakazato, Takashi Kawahara, Reiji Ichii, Shingo Akita, Akira Inoue, Masaki Fujita, Hideyuki Kurihara, Mitsuhiro Nakaya, and Atsushi Ooki Biogeosciences, 23, 2747–2759, <https://doi.org/10.5194/bg-23-2747-2026>, 2026 Seaweeds can store carbon in the ocean, but traditional methods in food chemistry fail to measure alginate—a major organic component derived from brown algae—in seafloor mud. We developed a novel extraction method for alginate in marine sediments. This allows for high recovery rates, and we successfully used this method to quantify alginate in marine sediments from northern Japan. Our work establishes a reliable method to track alginate on the ocean floor.

中文摘要

研究围绕“技术札记：海洋沉积物中海藻酸盐提取与比色分析方法”展开。英文摘要要点为：Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment Sota Nakazato, Takashi Kawahara, Reiji Ichii, Shingo Akita, Akira Inoue, Masaki Fujita, Hideyuki Kurihara, Mitsuhiro Nakaya, and Atsushi Ooki Biogeosciences, 23, 2747–2759, <https://doi.org/10.5194/bg-23-2747-2026>, 2026 Seaweeds can store carbon in the ocean, but traditional methods in food chemistry fail to measure alginate—a major organic component derived from brown algae—in seafloor mud. We developed a novel extraction method for alginate in marine sediments. This allows for high recovery rates, and we successfully used this method to quantify alginate in marine sediments from northern Japan. Our work establishes a reliable method to track alginate on the ocean floor. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2747-2026>

环境与气候

10. Decades of increased emissions from forest-fuelled BECCS

(以现有森林木材为燃料的 BECCS 可能在数十年内反而增加排放)

Nature Sustainability | 2026-04-20 | **【Recent】**

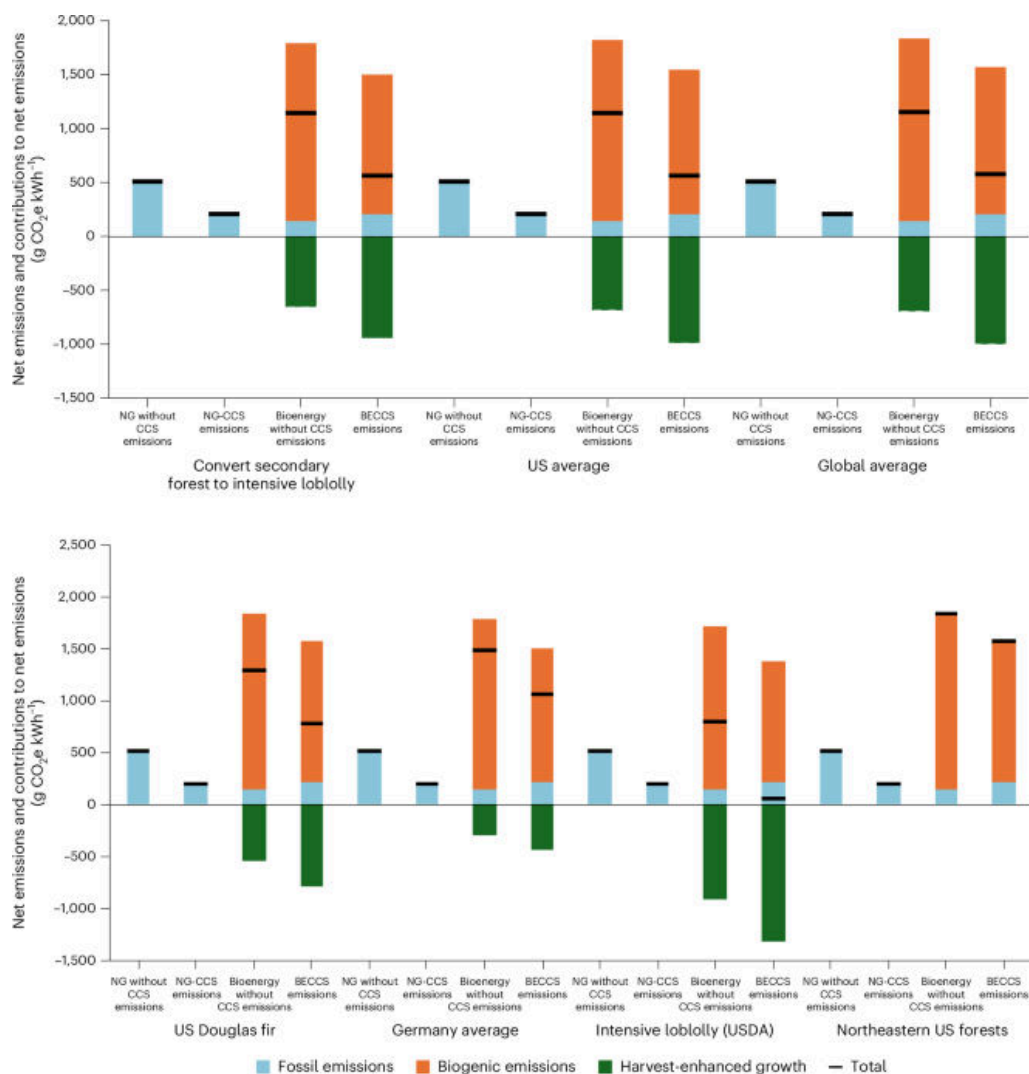


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 20 April 2026; doi:10.1038/s41893-026-01817-8 Governments are considering subsidies for bioenergy with carbon capture and storage fuelled by wood from existing forests. However, a transparent model estimates that this will probably increase emissions for decades, increase electricity costs 3.5-fold and not generate negative emissions for 150 years.

中文摘要

研究用透明模型评估以现有森林木材供能的生物质能碳捕集与封存（BECCS）方案，发现该路径可能在数十年内提高净排放、电力成本增加约 3.5 倍，并且约 150 年后才可能出现负排放效果。

深度解读

这篇文章直指负排放讨论中的一个痛点：并不是所有 BECCS 都天然减碳。若燃料来自现有森林，账面上的‘可再生’很可能掩盖了长期碳债，结果是前几十年排放更高、成本更贵、气候收益却极晚才出现。它的重要意义在于，把时间尺度重新拉回政策视野——对 2030 或 2050 目标来说，150 年后的收益几乎没有决策价值。对生物质补贴和碳中和路径设计，这是一记很实在的提醒。

DOI: <https://doi.org/10.1038/s41893-026-01817-8>

遥感与模型

11. Monitoring Coral Reef Metabolism Under Changing Oceans – Novel Insights From Seawater Stable Carbon Isotopes

(利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢)

Journal of Geophysical Research - Biogeosciences | 2026-04-18 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009416>

保护与生物多样性

12. Enhancement of climate-related plant translocations by reinforcing populations with nearby genotypes

(利用邻近基因型强化种群以提升气候相关植物迁地保护成效)

Conservation Biology | 2026-04-21 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“利用邻近基因型强化种群以提升气候相关植物迁地保护成效”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70289>

保护与生物多样性

13. Delayed plant genetic responses to forest edge dynamics and their conservation implications

(植物对森林边缘动态的遗传响应滞后及其保护启示)

Conservation Biology | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“植物对森林边缘动态的遗传响应滞后及其保护启示”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70297>

生态学核心

14. Multicellularity for free

(多细胞性演化的低成本路径)

Nature Ecology & Evolution | 2026-04-20 | 【Recent】

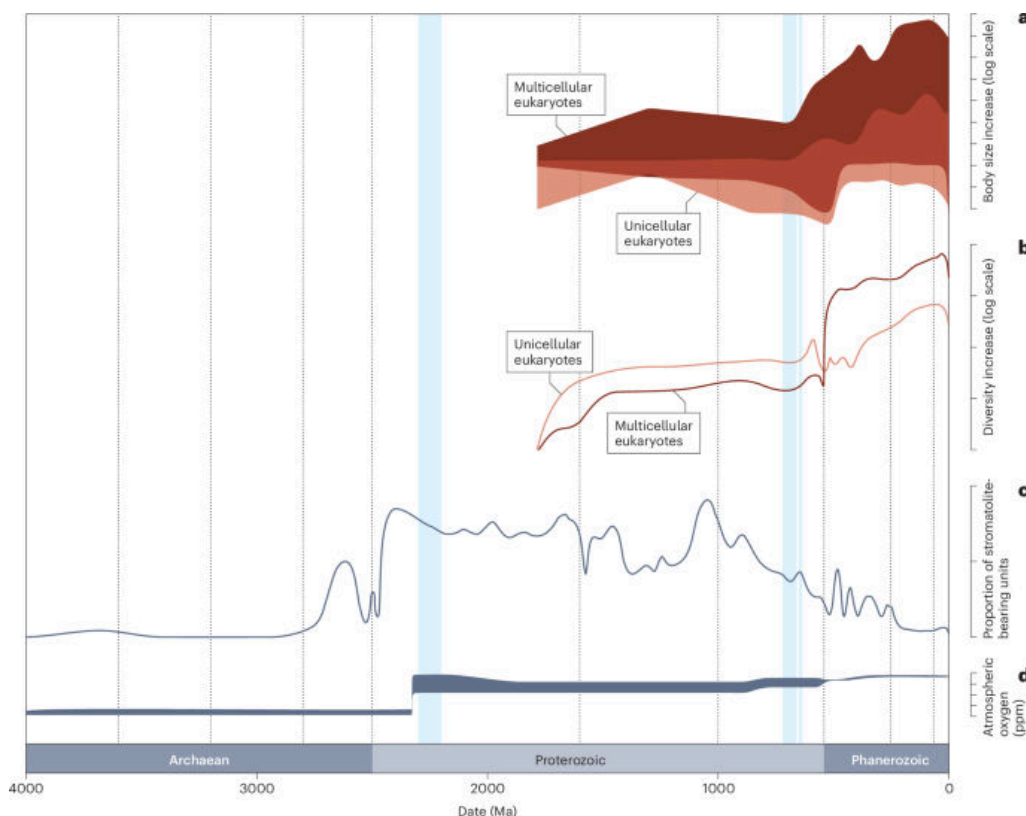


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03021-5A model of competition between unicellular and multicellular forms suggests that multicellularity can become ecologically stable without direct benefits, provided both forms use a varied environment in different ways.

中文摘要

研究围绕“多细胞性演化的低成本路径”展开。英文摘要要点为：A model of competition between unicellular and multicellular forms suggests that multicellularity can become ecologically stable without direct benefits, provided both forms use a varied environment in different ways. 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03021-5>

生态学核心

15. Direct benefits are not necessary for the evolution of multicellularity

(多细胞性演化并不必然需要直接收益)

Nature Ecology & Evolution | 2026-04-20 | 【Recent】

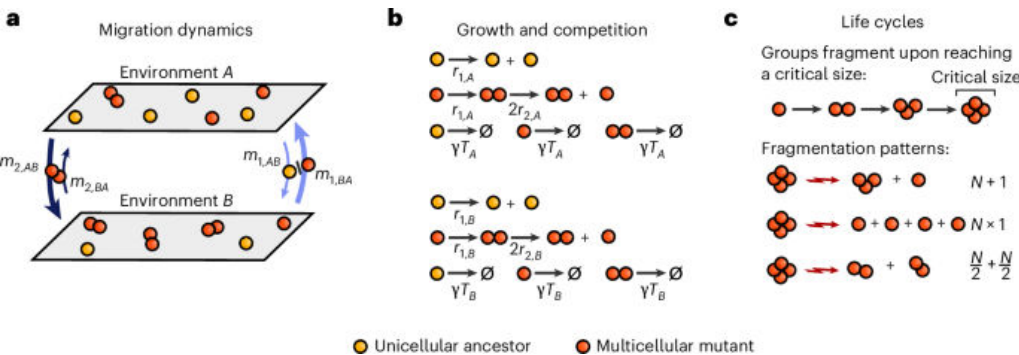


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03044-y A model for the evolution of multicellularity in a spatially heterogeneous environment shows that indirect mechanisms such as escape from competition and increased environmental exploitation can drive the evolution of multicellularity in the absence of direct benefits.

中文摘要

研究围绕“多细胞性演化并不必然需要直接收益”展开。英文摘要要点为：A model for the evolution of multicellularity in a spatially heterogeneous environment shows that indirect mechanisms such as escape from competition and increased environmental exploitation can drive the evolution of multicellularity in the absence of direct benefits. 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03044-y>

环境与气候

16. [ASAP] MarineTox Predictor: An Online Library Platform for Enhancing Low-Resourced Saltwater Ecotoxicity Prediction via Knowledge Sharing from Freshwater Ecotoxicity

(MarineTox Predictor: 借助淡水毒性知识提升海水生态毒性预测的在线平台)

Environmental Science & Technology | 2026-04-20 | 【Recent】



图: Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c15496

中文摘要

研究围绕“MarineTox Predictor: 借助淡水毒性知识提升海水生态毒性预测的在线平台”展开。英文摘要要点为: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c15496 该条目发表于 Environmental Science & Technology, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关, 能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c15496>

土壤与生物地球化学

17. Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods

(融合滑翔机观测与混合统计方法改进近岸海洋 pH 估算)

Biogeosciences | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods Jann Paul Mattern, Yuichiro Takeshita, Carlos Rocha, and Christopher A. Edwards Biogeosciences, 23, 2621–2639, <https://doi.org/10.5194/bg-23-2621-2026>, 2026 We improve coastal ocean carbonate system estimates by assimilating glider pH and alkalinity data into a regional biogeochemical model. Joint assimilation with physical observations successfully improves pH estimates while maintaining physical estimates. A hybrid approach combining dynamical models with statistical methods produces accurate pH estimates without requiring biogeochemical models, offering an alternative solution for ocean acidification monitoring.

中文摘要

研究围绕“融合滑翔机观测与混合统计方法改进近岸海洋 pH 估算”展开。英文摘要要点为：Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods Jann Paul Mattern, Yuichiro Takeshita, Carlos Rocha, and Christopher A. Edwards Biogeosciences, 23, 2621–2639, <https://doi.org/10.5194/bg-23-2621-2026>, 2026 We improve coastal ocean carbonate system estimates by assimilating glider pH and alkalinity data into a regional biogeochemical model. Joint assimilation with physical observations successfully improves pH estimates while maintaining physical estimates. A hybrid approach combining dynamical models with statistical methods produces accurate pH estimates without requiring biogeochemical models, offering an alternative solution for ocean acidification monitoring. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2621-2026>

保护与生物多样性

18. Relative importance of traits, climate, and threats to extinction risk in salamanders

(性状、气候与威胁因素对蝾螈灭绝风险的相对重要性)

Conservation Biology | 2026-04-19 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“性状、气候与威胁因素对蝾螈灭绝风险的相对重要性”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70281>

植物与农业

19. Turning Mediterranean Farmlands Into Priority Habitats: Natural Expansion of Juniper Woodlands After Agricultural Abandonment

(农业废弃后刺柏林自然扩张将地中海农田转变为优先栖息地)

Journal of Vegetation Science | 2026-05-02 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 3, May/June 2026.

中文摘要

研究关注地中海农田废弃后刺柏林自然扩张的生态后果，指出部分废弃农田可逐步转化为具有保护价值的优先栖息地。该结果说明，土地利用退出并不总是退化过程，在合适区域也可能促进半自然木本群落恢复。

深度解读

地中海地区的农业废弃常被视为社会经济问题，但生态结果取决于恢复路径和目标物种。刺柏林自然扩张显示，被动恢复在某些景观中可以形成有保护价值的栖息地。该研究对欧洲和半干旱地区土地利用转型具有启发：不是所有废弃地都需要高强度造林或工程修复。关键是识别哪些地块适合自然恢复，哪些地块需要主动管理以避免火险或单一化。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70142>

遥感与模型

20. Reducing Uncertainties in Net Carbon Capture to Advance Wetlands as Natural Climate Solutions

(降低净碳捕获不确定性以推动湿地成为自然气候解决方案)

Journal of Geophysical Research - Biogeosciences | 2026-04-20 | **【Recent】**



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“降低净碳捕获不确定性以推动湿地成为自然气候解决方案”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009136>

土壤与生物地球化学

21. Organic Carbon and Texture Control Moisture Dependence of Soil Shortwave Infrared Reflectance (有机碳和质地控制土壤短波红外反射率的水分依赖性)

European Journal of Soil Science | 2026-04-19 | 【Recent】



图： Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“有机碳和质地控制土壤短波红外反射率的水分依赖性”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70319>

生态学核心

22. Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO₂, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model

(CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟)

Global Change Biology | 2026-04-18 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟”展开。英文摘要要点为：Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO₂, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70867>

生态学核心

23. Fast range shifts

(物种分布范围的快速迁移)

Nature Ecology & Evolution | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“物种分布范围的快速迁移”展开。英文摘要要点为：Fast range shifts 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03072-8>

环境与气候

24. [ASAP] Free Amino Acids in Marine Aerosols during Four Cruising Campaigns over the South China Sea: Importance of Biomass Burning and Atmospheric Oxidation

(南海航次海洋气溶胶游离氨基酸：生物质燃烧与大气氧化的重要作用)

Environmental Science & Technology | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c18403

中文摘要

研究围绕“南海航次海洋气溶胶游离氨基酸：生物质燃烧与大气氧化的重要作用”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.5c18403 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c18403>

环境与气候

25. [ASAP] Fe(III) Oxide Reduction Bypassing Outer-Surface Cytochromes in a Marine Respiratory Anaerobe (海洋呼吸厌氧菌绕过外表面细胞色素还原三价铁氧化物)

Environmental Science & Technology | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.6c04532

中文摘要

研究围绕“海洋呼吸厌氧菌绕过外表面细胞色素还原三价铁氧化物”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.6c04532 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.6c04532>

生态学核心

26. Estimating the Impacts of Future Extreme Heat on Dryland Threatened Mammals: An Australian Case Study (估算未来极端高温对干旱区受威胁哺乳动物的影响：澳大利亚案例)

Global Change Biology | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“估算未来极端高温对于干旱区受威胁哺乳动物的影响：澳大利亚案例”展开。英文摘要要点为：Estimating the Impacts of Future Extreme Heat on Dryland Threatened Mammals: An Australian Case Study 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70872>

植物与农业

27. Membrane remodelling mediates lipopeptide-induced immunity in Arabidopsis

(膜重塑介导拟南芥中脂肽诱导的免疫反应)

Nature Plants | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 20 April 2026; doi:10.1038/s41477-026-02270-3 This work unveils a non-canonical lipid-driven mechanism for plant immune activation by a bacterial lipopeptide. Perception at the cell membrane leads to deformation, mechanosensing and early signalling and culminates in systemic resistance priming.

中文摘要

研究围绕“膜重塑介导拟南芥中脂肽诱导的免疫反应”展开。英文摘要要点为：This work unveils a non-canonical lipid-driven mechanism for plant immune activation by a bacterial lipopeptide. Perception at the cell membrane leads to deformation, mechanosensing and early signalling and culminates in systemic resistance priming. 该条目发表于 Nature Plants，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41477-026-02270-3>

土壤与生物地球化学

28. Phosphogypsum Doses and Subsoil Contamination: A Study on the Mobility of Rare Earth Elements, Uranium, and Thorium

(磷石膏施用量与底土污染：稀土元素、铀和钍迁移性研究)

European Journal of Soil Science | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“磷石膏施用量与底土污染：稀土元素、铀和钍迁移性研究”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

保护与生物多样性

29. Understanding drivers of coastal fishers' willingness to exit fishing in Zhejiang, China
(理解中国浙江沿海渔民退出捕捞意愿的驱动因素)

Conservation Biology | 2026-04-19 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“理解中国浙江沿海渔民退出捕捞意愿的驱动因素”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70294>

生态学核心

30. Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions

(放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收)

Global Change Biology | 2026-04-18 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收”展开。英文摘要要点为：Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70864>

综合顶刊

31. Historic logbooks reveal spatial footprints of commercial whaling

(历史航海日志揭示商业捕鲸的空间足迹)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

SignificanceCommercial whaling devastated bowhead whale populations, and recovery has been slow and spatially heterogeneous. By reconstructing daily routes and hunting activities of more than 700 historic British and American whaling voyages, we mapped ...

中文摘要

研究利用 700 多次英美商业捕鲸航行的历史日志，重建捕鲸路线和猎捕活动的空间足迹。结果表明，商业捕鲸对弓头鲸等鲸类种群造成深刻影响，且恢复过程存在明显空间异质性，历史捕捞压力需要纳入当代海洋保护评估。

深度解读

历史生态数据常被低估，但它们能揭示现代监测开始前已经发生的强烈人类影响。商业捕鲸的空间足迹不仅解释了鲸类恢复不均，也会影响海洋碳循环、营养盐再分配和食物网结构。该研究提示海洋保护目标应纳入历史捕捞压力和空间记忆，避免把已经贫化的海洋状态当作自然基线。对生态修复而言，历史文献数据是重建长期基线的重要证据来源。

DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2523917123>

综合顶刊

32. Repeated snowball–hothouse cycles within the Neoproterozoic Sturtian glaciation

(新元古代斯图尔特冰期内反复出现雪球-温室循环)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

SignificanceEarth experienced extreme climate swings during the Neoproterozoic epoch, including the Sturtian glaciation, when ice likely covered the planet. Explaining aspects of the geologic record and the survival of life through this event has been a ...

中文摘要

研究围绕新元古代斯图尔特冰期的极端气候波动展开，讨论冰封地球状态下雪球与温室条件可能反复交替。该结果有助于解释地质记录中的异常信号，也为理解极端气候转换期间生命如何持续提供背景。

深度解读

这项研究虽然偏古气候，但对生态环境速览有价值，因为它揭示地球系统可在极端状态之间发生反复跃迁。极端气候循环会改变风化、海洋化学和生物可利用栖息地，对生命演化背景具有根本影响。它提醒我们，气候系统存在非线性阈值和反馈，不能简单按线性外推理解长期生态风险。现代气候变化尺度不同，但阈值思维仍值得借鉴。

海洋与水生

33. A bioinspired hierarchical architecture for the high-yield recovery of industrial water vapour
(仿生层级结构实现工业水蒸气高产回收)

Nature Water | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Nature Water, Published online: 21 April 2026; doi:10.1038/s44221-026-00635-8 Industrial cooling towers release large quantities of water vapour that remains largely unutilized, despite its potential as a recoverable resource. This study demonstrates a passive, biomimetic multi-tier system to achieve high-rate water recovery with substantial large-scale savings potential.

中文摘要

研究围绕“仿生层级结构实现工业水蒸气高产回收”展开。英文摘要要点为：Industrial cooling towers release large quantities of water vapour that remains largely unutilized, despite its potential as a recoverable resource. This study demonstrates a passive, biomimetic multi-tier system to achieve high-rate water recovery with substantial large-scale savings potential. 该条目发表于 Nature Water，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s44221-026-00635-8>

环境与气候

34. [ASAP] Effects of Pesticide Mixtures and Environmental Factors on Benthic Diatom Communities: A Microcosm Approach

(农药混合物与环境因素对底栖硅藻群落的影响：微宇宙实验)



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c14415

中文摘要

研究围绕“农药混合物与环境因素对底栖硅藻群落的影响：微宇宙实验”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.5c14415 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c14415>

土壤与生物地球化学

35. Composite Soil Samples Reflect the Arithmetic Mean pH of the Individual Cores

(混合土样可反映单个土芯 pH 的算术平均值)

European Journal of Soil Science | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“混合土样可反映单个土芯 pH 的算术平均值”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70329>

遥感与模型

36. Spatiotemporal Modulation of Alkalinity and DIC Outwelling From Saltmarsh Porewater in New England's Largest Marsh Complex

(新英格兰最大盐沼复合体孔隙水碱度与溶解无机碳外输的时空调制)

Journal of Geophysical Research - Biogeosciences | 2026-04-20 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“新英格兰最大盐沼复合体孔隙水碱度与溶解无机碳外输的时空调制”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009711>

保护与生物多样性

37. Integrating traditional practices, livelihoods, and conservation with Indigenous-led furbearer camera trapping
(以原住民主导的毛皮动物红外相机监测整合传统实践、生计与保护)

Conservation Biology | 2026-04-19 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“以原住民主导的毛皮动物红外相机监测整合传统实践、生计与保护”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70290>

土壤与生物地球化学

38. Depth and Tillage Dependent Stoichiometry of C, N, S, and H in Ukrainian Mollisol
(乌克兰黑钙土碳氮硫氢化学计量随深度和耕作方式变化)

European Journal of Soil Science | 2026-04-19 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“乌克兰黑钙土碳氮硫氢化学计量随深度和耕作方式变化”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70335>

植物与农业

39. Process-based modelling in farm greenhouse gas assessment reveals site-specific dynamics and limitations of emission factor methods

(农场温室气体评估中的过程模型揭示地点特异动态和排放因子法局限)

Agronomy for Sustainable Development | 2026-04-21 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Life cycle assessment (LCA) is a well-recognized tool to assess the environmental impact of food production. To assess partial life-cycle greenhouse gas emissions in agricultural systems with variable weather and management, this study integrated process-based agricultural systems modelling (APSIM) with emission-factor-based calculations to develop a system modelling framework. We assessed the greenhouse gas emission intensity of 11 fields from 2016 to 2021 at Boorowa Agricultural Research Station, representing an Australian cropping farm with comprehensive management records. Net greenhouse gas emissions varied widely across fields and seasons, ranging from -3.87 to $6.10 \text{ t CO}_2\text{-e ha}^{-1}$. Emissions were not only determined by seasonal climate but also prior-year management decisions, highligh

中文摘要

研究将 APSIM 等过程农业系统模型与排放因子计算结合,用于评估澳大利亚农场多个田块 2016-2021 年温室气体排放强度。结果显示,净排放在田块和季节之间差异很大,受天气、作物序列和前一年管理共同影响,固定排放因子方法容易低估地点特异动态。

深度解读

农田减排政策常依赖排放因子,但这种方法难以捕捉天气、土壤和管理历史造成的年际差异。过程模型可以把管理措施与生物地球化学过程连接起来,更适合做地点化评估。该研究提示低碳农业不能只给出统一系数,而应建立田块尺度的动态核算框架。对农业生态系统管理来说,这有助于识别真正有效的减排组合。

DOI: <https://link.springer.com/article/10.1007/s13593-026-01104-y>

植物与农业

40. Crop Rotation Index for measuring agricultural land use intensity

(用于衡量农业土地利用强度的作物轮作指数)

Agronomy for Sustainable Development | 2026-04-30 | 【Latest】

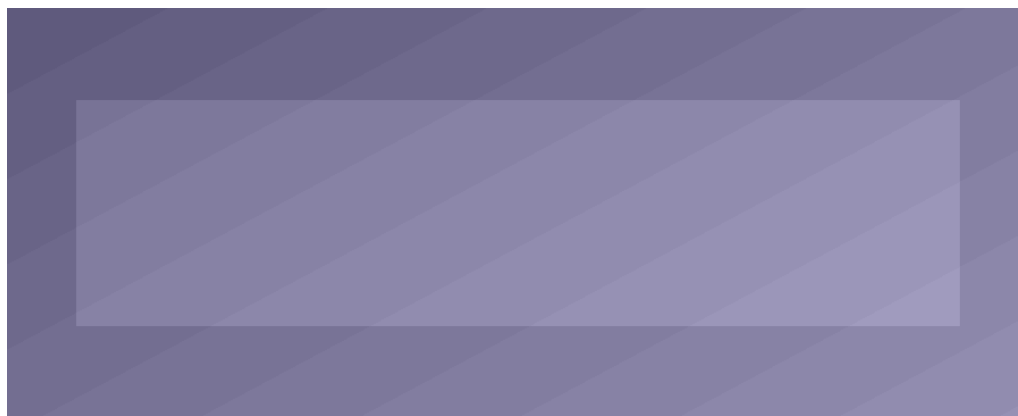


图: Graphical abstract / article figure source:

Abstract

Crop rotations constitute a crucial way of managing farms. Scientific methods have studied rotations through different approaches. These either focus on the design of rotations, the performance of individual rotations (from a productivity, environmental performance, and/or soil recovery point of view), or the spatial pattern in the landscape. Despite these approaches, there is no method available to investigate the sustainability of a rotation at a large scale based on consistently followed agronomic principles in everyday farming practices. The novel Crop Rotation Index is proposed to fill this gap. To compute the Crop Rotation Index on a parcel basis, a sequence of crops over 6 years is evaluated by 8 indicators, where crops in the sequence are grouped together based on botanical familie

中文摘要

研究提出作物轮作指数，用 6 年作物序列和 8 个农艺指标在地块尺度评估农业土地利用强度。该方法试图把轮作多样性、植物科属组合和管理原则转化为可在大尺度农业景观中比较的量化指标。

深度解读

轮作是连接产量、土壤健康、病虫害压力和生物多样性的关键管理措施，但大尺度量化一直困难。作物轮作指数把复杂的轮作序列压缩成可比较指标，为遥感、农情统计和可持续农业评估提供接口。虽然该条目评分较低，但它补齐了当周农业土地利用强度方法学方向。后续需要验证该指数是否能稳定预测土壤碳、农药使用和景观生物多样性结果。

DOI: <https://link.springer.com/article/10.1007/s13593-026-01101-1>
